

MELBOURNE ESSENDON, AustraliaWMO: **958660**Lat: **37.7275S**Lon: **144.9067E**Elev: **259**StdP: **14.56**Time Zone: **10.00 (AUS)**Period: **03-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.4	39.5	32.1	26.7	57.4	34.0	28.9	52.5	29.3	54.0	27.1	53.5	5.3	0	0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.8	96.1	66.3	91.1	65.3	86.5	64.5	70.5	84.3	68.7	82.7	67.2	80.5	15.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.9	100.0	73.6	64.7	92.3	72.0	63.0	87.0	70.9	34.6	85.6	33.1	83.2	31.8	80.5	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.2	23.5	21.2	32.6	106.4	1.8	4.1	31.3	109.3	30.3	111.7	29.3	114.1	28.0	117.1		
(5)			32.1	74.0	1.7	1.9	30.8	75.4	29.8	76.5	28.9	77.6	27.6	79.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.2	69.2	68.5	66.0	60.3	55.0	50.7	49.9	50.9	54.1	58.2	62.6	65.8		
	DBStd	8.88	7.47	6.32	6.50	5.46	4.34	3.50	3.29	3.92	4.75	6.21	7.10	7.04		
	HDD50	125	0	0	0	0	5	33	43	34	8	2	0	0		
	HDD65	2701	31	30	65	162	312	430	468	437	329	231	133	73		
	CDD50	3486	594	518	497	311	159	52	40	62	132	255	377	489		
	CDD65	585	160	128	97	22	0	0	0	0	3	19	59	97		
	CDH74	6394	1784	1227	935	191	1	0	0	0	18	286	736	1216		
	CDH80	2847	926	529	383	36	0	0	0	0	2	84	290	597		
(14) Wind	WSAvg	10.9	11.5	10.9	10.4	9.5	10.2	9.6	11.5	11.5	11.6	11.5	11.3	11.3		
(15) Precipitation	PrecAvg	21.1	1.7	1.7	1.3	1.8	1.6	1.5	1.4	1.8	1.9	2.2	2.3	1.9		
	PrecMax	32.3	4.0	7.9	3.4	5.6	6.1	4.1	3.7	3.8	5.0	5.7	6.2	5.5		
	PrecMin	12.2	0.2	0.0	0.2	0.2	0.3	0.5	0.3	0.6	0.4	0.6	0.7	0.1		
	PrecStd	4.8	0.9	1.9	0.7	1.3	1.1	0.8	0.8	0.8	1.0	1.3	1.4	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	105.6	97.7	96.6	84.5	72.0	63.7	62.1	68.0	76.1	88.0	93.1	101.1		
		MCWB	69.0	67.2	66.8	61.2	57.0	54.3	53.5	56.1	58.4	60.5	63.9	65.9		
	2%	DB	96.9	91.8	89.0	79.3	68.1	60.7	59.4	63.8	70.7	81.4	87.9	93.7		
		MCWB	66.4	66.4	64.6	60.2	57.1	53.3	52.2	53.5	57.1	60.0	63.5	64.9		
	5%	DB	90.2	86.6	83.8	75.1	65.3	58.8	57.6	61.0	67.1	76.3	82.5	86.4		
		MCWB	66.3	65.8	63.8	59.6	56.1	52.5	50.9	51.9	55.3	58.7	62.8	63.5		
	10%	DB	83.6	81.4	78.5	71.4	62.9	56.9	55.9	58.3	63.9	71.3	77.1	80.2		
		MCWB	65.2	65.6	63.0	58.6	54.8	51.7	49.9	50.6	53.4	57.2	62.2	63.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.1	72.0	70.2	65.0	61.2	56.6	55.8	56.9	60.5	64.9	69.8	71.1		
		MCDB	85.4	85.8	83.4	77.5	67.4	59.9	60.2	66.4	71.0	77.7	83.9	84.5		
	2%	WB	70.8	69.9	67.9	62.9	58.5	54.8	53.4	54.3	58.2	62.1	66.4	67.9		
		MCDB	85.8	82.4	80.5	72.1	65.2	58.5	57.5	62.0	68.3	75.8	78.3	82.7		
	5%	WB	68.8	68.2	66.3	61.4	57.1	53.6	51.8	52.6	56.3	60.0	64.8	66.0		
		MCDB	84.0	80.8	76.9	70.1	63.5	57.9	56.1	59.2	65.5	72.7	77.0	80.0		
	10%	WB	66.8	66.6	64.8	59.8	55.6	52.3	50.6	51.3	54.3	57.9	62.7	64.3		
		MCDB	81.5	78.5	75.3	68.8	61.3	56.3	54.9	57.3	62.3	69.3	75.7	77.4		
(35) Mean Daily Temperature Range	MDBR		19.8	18.7	18.2	16.8	13.4	11.8	11.3	13.1	16.0	18.9	19.0	20.5		
	5% DB	MCDBR	34.1	31.3	28.8	24.7	16.8	14.6	13.0	16.5	21.5	29.3	30.4	34.5		
		MCWBR	9.7	10.0	9.8	10.2	9.1	9.0	7.9	8.9	11.2	12.2	10.4	10.9		
	5% WB	MCDBR	25.7	22.9	20.9	19.1	14.3	12.2	11.0	15.0	19.2	24.8	24.1	25.5		
		MCWBR	9.5	9.5	8.8	9.2	8.6	8.1	7.4	8.8	10.6	11.8	10.0	10.1		
(40) Clear-Sky Solar Irradiance	taub		0.349	0.354	0.329	0.330	0.311	0.301	0.301	0.313	0.332	0.335	0.347	0.340		
	taud		2.509	2.503	2.572	2.537	2.574	2.592	2.583	2.534	2.478	2.491	2.484	2.523		
	Ebn at Noon		312	303	298	278	265	259	266	279	292	306	311	316		
	Edh at Noon		36	34	30	27	22	20	22	26	32	34	36	35		
(44) All-Sky Solar Radiation	RadAvg		2274	1984	1584	1099	727	563	628	889	1269	1695	1997	2238		
	RadStd		144	104	91	65	44	45	37	61	92	137	121	126		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (39 neighbors)	+0.90	N/A	N/A	+1.57	N/A	N/A	-31	-229	+306	+104

Nomenclature: See separate page